PROJECT REPORT

English School Management Platform

Sprint 0 – Analysis & Specification

Prepared by
Akrem Agroubi
Firas Tourki
Youssef Feddaoui
Rayen Karouch
Kamel Hamdi

Supervisor
Nadine Mili

ESPRIT – Honoris United Universities
Academic Year 2025–2026

Abstract

Language schools manage complex academic and administrative processes including students, teachers, courses, schedules, assessments, and payments. Many institutions still rely on manual or fragmented systems, leading to inefficiencies and data inconsistencies.

This project aims to design an English School Management Platform based on a modern and scalable architecture. The proposed solution adopts a microservices-based backend using Spring Boot and a web frontend developed with Angular. This report presents Sprint 0, which focuses on requirements analysis, system modeling, architectural design, and project organization.

General Introduction

The rapid evolution of digital technologies has significantly impacted the education sector. Language schools must manage various academic and organizational activities that require efficient information systems. However, existing solutions are often costly, complex, and not adapted to the specific needs of language education.

This project proposes a centralized and modular platform that automates academic and administrative workflows while ensuring scalability, security, and maintainability.

Contents

Abstract **1**

General Introduction **2**

1 Project Context and Study of Existing Solutions **4**

1.1 Project Context 4

1.2 Existing Solutions 4

1.3 Limitations of Existing Solutions 4

1.4 Proposed Solution 4

2 Requirements Analysis **5**

2.1 Functional Requirements 5

2.2 Non-Functional Requirements 5

3 System Modeling **6**

3.1 Use Case Diagram 6

3.2 Class Diagram 7

4 System Architecture **8**

4.1 Global Architecture 8

4.2 Logical Architecture 8
4.3 Physical Architecture 9

5 Project Organization and Management 11
5.1 Microservices Distribution 11
5.2 Methodology and Tools 11
5.3 Project Roadmap 11

General Conclusion 12

Chapter 1. Project Context and Study of Existing Solutions

1.1 Project Context

Language schools manage students, teachers, courses, schedules, exams, and payments. Manual and semi-digital processes cause inefficiency and lack of data consistency.

1.2 Existing Solutions

Several solutions exist such as MyScol, LGE (Logiciel de Gestion d'Établissement), and generic ERP systems for schools.

1.3 Limitations of Existing Solutions

These systems often suffer from high licensing costs, limited customization, complex interfaces, and lack of specialization for language schools.

1.4 Proposed Solution

To overcome these limitations, we propose a web-based platform based on microservices, designed specifically for language schools.

Chapter 2. Requirements Analysis

2.1 Functional Requirements

The platform provides the following functionalities:

- User authentication and role management
- Course and enrollment management
- Scheduling and attendance tracking
- Exams, grades, and learning resources
- Timed quizzes with score comparison
- Payments, reporting, and notifications

2.2 Non-Functional Requirements

- Performance and responsiveness
- Security and data protection
- Scalability and availability
- Usability and maintainability

Chapter 3. System Modeling

3.1 Use Case Diagram

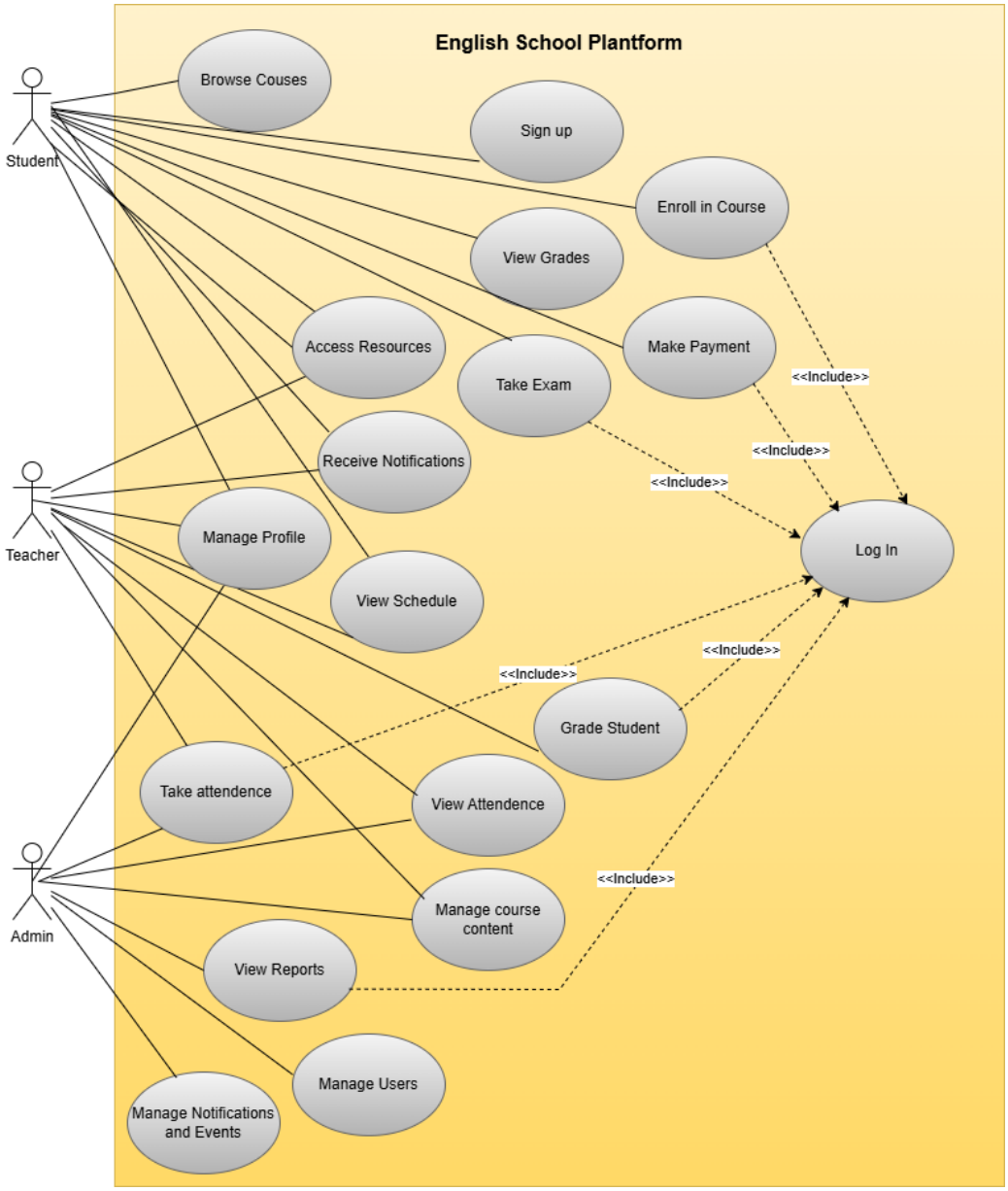

Figure 3.1: Use Case Diagram

3.2 Class Diagram

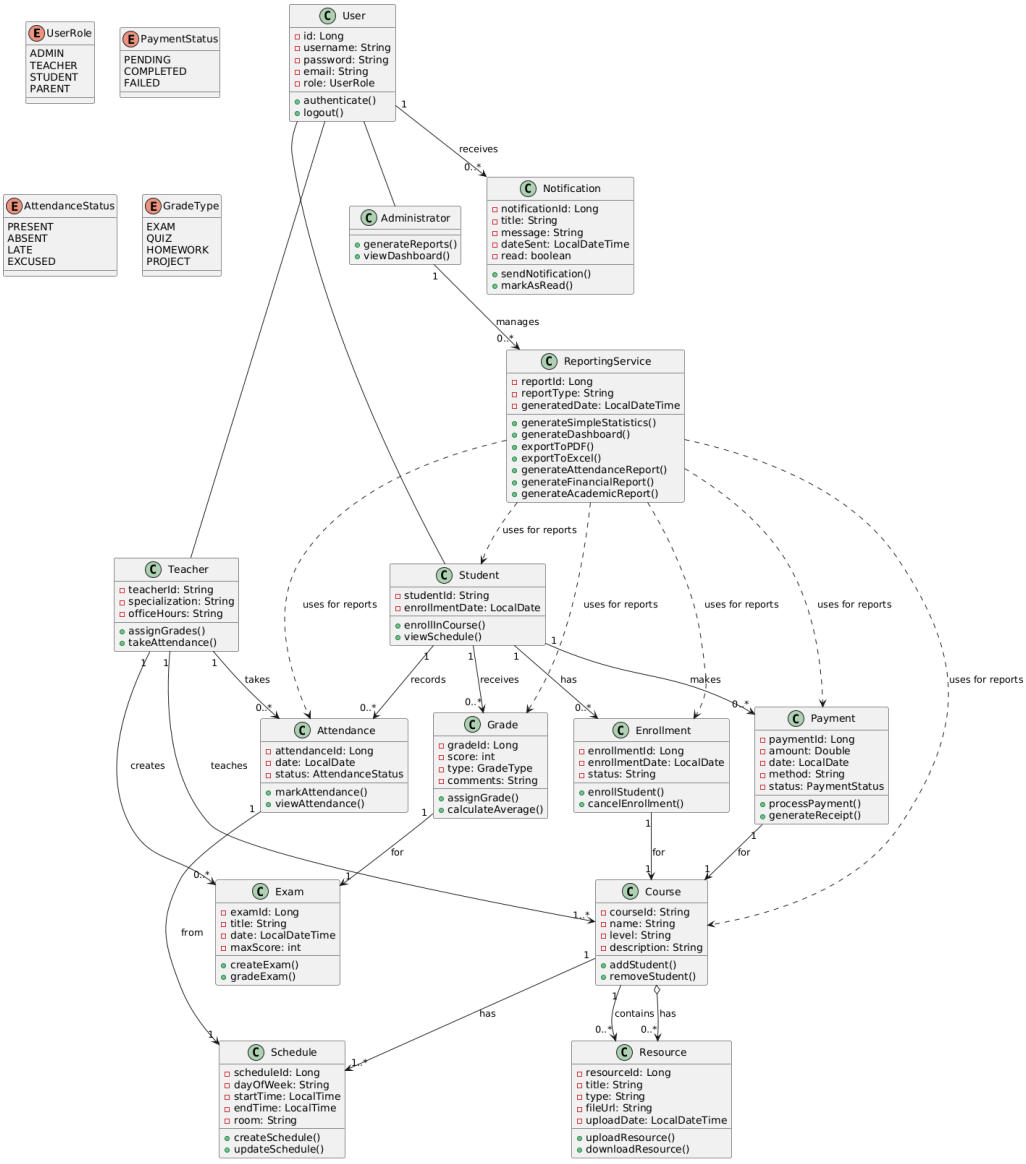

Figure 3.2: Class Diagram

Chapter 4. System Architecture

4.1 Global Architecture

The platform follows a three-tier architecture composed of a presentation layer, an application layer, and a data layer.

4.2 Logical Architecture

The logical architecture is based on independent microservices communicating through REST APIs.

Figure 4.1: Logical Architecture of the English School Management Platform

4.3 Physical Architecture

The physical architecture describes the deployment infrastructure of the platform. Each microservice is containerized and deployed independently to ensure scalability and fault isolation.

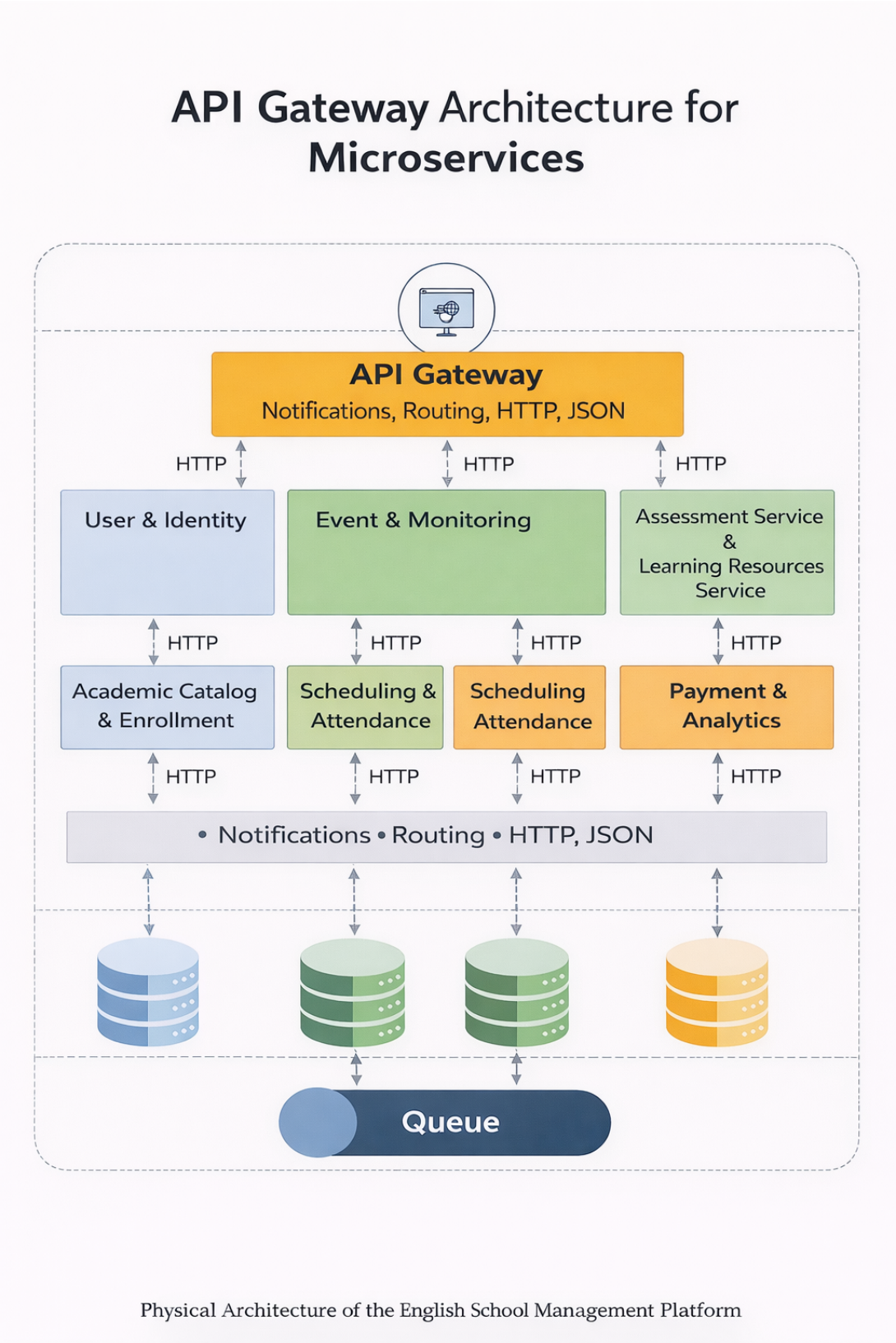

Figure 4.2: Physical Architecture of the English School Management Platform

Chapter 5. Project Organization and Management

5.1 Microservices Distribution

- Akrem Agroubi: User & Identity, Event & Monitoring
- Firas Tourki: Assessment Service & Learning Resources Service
- Youssef Feddaoui: Course Service & Enrollment Service
- Rayen Karouch: Scheduling Service & Attendance Service
- Kamel Hamdi: Payment Service & Analytics Service

5.2 Methodology and Tools

The project follows an Agile Scrum methodology using Jira for task management and GitHub for version control. Technologies include Spring Boot, Angular, MySQL, and Docker.

5.3 Project Roadmap

- Sprint 1–2: Core microservices development
- Sprint 3: DevOps and deployment
- Sprint 4: Analytics and advanced features

General Conclusion

This report presented the analysis and specification phase of the English School Management Platform. Sprint 0 established a clear functional and technical vision based on a microservices architecture. The defined requirements, UML models, and system architecture provide a solid foundation for the implementation phase in upcoming sprints.